AI & ML Milestones for Insights Teams, 1950–2024

What each milestone means for research and brand teams

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FOUNDATIONS · 1950–1966

1950 — **Turing asks whether machines can think**
We still judge AI output by whether it sounds human, for example AI-written open-end summaries or synthetic respondents. Sounding human is not the same as being accurate, so the output still has to be checked against real data.

1955–56 — **The Dartmouth proposal names the field**
That early optimism still shows up in vendor pitches. Knowing how long real progress took helps a research team ask what a tool can actually do today.

1958 — **Rosenblatt's perceptron learns from examples**
This is the basic idea behind automated coding of survey answers: show the model coded examples and let it learn the pattern.

1959 — **Samuel's checkers program improves with practice**
It is an early case of a computer learning from experience, the same logic as a model that improves as more survey waves or transactions come in.

1966 — **ELIZA, the first well-known chatbot**
A direct warning for chatbot-led interviews and AI moderators: a fluent reply does not mean the system understood the respondent.

LEARNING FROM DATA · 1986–2001

1986 — **Backpropagation makes deep networks trainable**
Almost every neural model used in text and image analytics today is trained this way.

2001 — **Random forests**
For driver-style questions (what is linked to satisfaction, switching or purchase intent), this gives analysts a model that handles messy survey data and still shows which variables carry weight.

DEEP LEARNING · 2012–2017

2012 — **Deep learning wins ImageNet**
Image recognition is now used to read shelf photos, packaging and ad creative, which used to need manual review.

2013 — **Word2vec turns words into numbers**
This made it practical to group open-ended answers by meaning, not just exact keywords, for example "too pricey" and "expensive."

2017 — **The transformer architecture**
Transformers are the foundation of today's large language models; the "T" in GPT stands for transformer.

GENERATIVE AI AND REGULATION · 2022–2024

2022 — **ChatGPT is released to the public**
Large language models moved from labs to everyone's desk: analysts, clients and respondents. Expectations for how fast insights can be produced changed almost overnight.

2024 — **The EU Artificial Intelligence Act**
Teams using AI with EU customers or data need to know which risk category their use falls into and what transparency it requires.

Sources: Turing (1950); McCarthy et al. (1955/2006); Rosenblatt (1958); Samuel (1959); Weizenbaum (1966); Rumelhart et al. (1986); Breiman (2001); Krizhevsky et al. (2012); Mikolov et al. (2013); Vaswani et al. (2017); OpenAI (2022); Regulation (EU) 2024/1689 (2024). Full APA 7 references in the online version of this artifact.